CLASS 10 - MATHEMATICS

Sample Student Answer Sheet

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Q1

36 = 2 x 2 x 3 x 3

48 = 2 x 2 x 2 x 2 x 3

HCF = 12

Q2

3x - 7 = 11

3x = 18

x = 6

Q3(a)

Distance = sqrt((5-2)^2 + (7-3)^2)

= sqrt(9+16) = 5 units

Q4

Prime numbers are 2 and 5.

Probability = 2/6 = 1/3

Q5

Circumference = 2 pi r
= 2 x 22/7 x 7 = 44 cm

Q6

x + y = 27
x - y = 5
2x = 32, x = 16
y = 11

Q7

Using Pythagoras theorem:
base^2 = 10^2 - 8^2
= 100 - 64 = 46
base = sqrt(46) = 6.78 m

Q8

Mean = (6+8+10+12+14+16+18)/7
= 84/7 = 12

Q9

CSA = 2 pi r h
= 2 x 22/7 x 3.5 x 10
= 220 cm^2

Q10(a)

x^2 - 7x + 12 = 0
(x - 3)(x - 4)=0
x = 3, 4

Q10(b)

a = 5, d = 3
20th term = a + 20d
= 5 + 60 = 65

Q11

Outer area = 600 m^2
Inner area = 26 x 16 = 416 m^2
Path area = 184 m^2
Cost = 184 x 100 = Rs. 18,400